

Smart Parking System

Version 1.0

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Documentation

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1 Introduction

1.1 Overview

The **Smart Parking System** is designed to optimize urban parking management with advanced sensors, IoT integration, and modern software platforms. It enhances parking efficiency by providing real-time data, automation, and intuitive user experiences.

1.2 Purpose Scope

This document guides stakeholders, developers, integrators, and users to understand, deploy, and maintain the Smart Parking System.

1.3 Key Features

- Real-time spot detection
- Reservation and payment integration
- User notifications
- Web/mobile interfaces
- Reporting and analytics

2 System Architecture

2.1 Hardware Components

- **Sensors:** Vehicle detection
- **Cameras:** Plate recognition, security
- **Gates/Barriers:** Automated access

Figure 1: System Architecture Diagram (Placeholder)

2.2 Software Modules

- **Mobile App:** Spot search, reservation, payment
- **Web Dashboard:** Admin analytics and control
- **Backend Server:** Core logic and data processing
- **Database:** User, spot, and transaction storage

2.3 Communication Protocols

- MQTT/HTTP for device-server communication
- REST API/GraphQL for client-server
- WebSockets for real-time updates

3 Functional Requirements

- Spot detection and status
- Real-time availability display
- Reservation features
- Payment integration
- User notifications
- Reporting and analytics

4 Non-Functional Requirements

- **Performance:** Response time < 2s
- **Security:** SSL/TLS, authentication
- **Scalability:** 10,000+ concurrent spots
- **Usability:** Accessible UI
- **Reliability:** $\geq 99.9\%$ uptime

5 User Guides

5.1 For Drivers

1. Registration
2. Find and reserve a spot
3. Navigate to spot
4. Make payment
5. Receive notifications

5.2 For Administrators

1. Dashboard login
2. Manage areas/spots
3. Monitor usage and revenue
4. Export reports
5. Manage users/permissions

6 API Documentation

6.1 Authentication

POST /api/auth/login

```
{ "username": "user", "password": "hashed_password" }
```

6.2 Endpoints

Method	Endpoint	Description	Auth
GET	/api/spots	List available spots	Yes
POST	/api/reservations	Reserve spot	Yes
POST	/api/payments	Payment processing	Yes
GET	/api/reports/usage	Analytics	Admin

6.3 Request/Response Examples

POST /api/reservations

```
{ "user_id": "123", "spot_id": "A10", "start_time": "2025-11-18T09:00" }
```

Response:

```
{ "reservation_id": "456", "status": "confirmed" }
```

7 System Deployment

- Hardware installation (sensors, cameras, gates)
- Network integration
- Backend/server deployment
- Software configuration
- Testing and diagnostics

8 Maintenance & Troubleshooting

8.1 Common Issues

- Device offline
- Payment gateway errors
- Login failure

8.2 Solutions

- Reboot devices
- Check API keys
- Password reset

8.3 Contact Support

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9 Appendix & References

9.1 Glossary

- **REST API:** Representational State Transfer API
- **SSL:** Secure Sockets Layer
- **TLS:** Transport Layer Security
- **WCAG:** Web Content Accessibility Guidelines

9.2 Related Resources

Open Parking Standards, Urban Mobility Guidelines